Reinforcement Learning Assignment

PROJECT 23: ANIMAL FORAGING BEHAVIOR USING Q-LEARNING

(Project category: RL in biological systems)

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PROBLEM FORMULATION

Motivation : Animals must decide whether to remain in a food patch or move elsewhere as food is depleted and replenished. We formulate this decision as a finite-horizon Markov decision process (MDP). The aim is to maximize food reward minus movement costs.

Environment. : A simulated animal occupies one of two patches, A or B.

Each patch has 0–3 food units. Patch A yields,
2.2 reward per collected unit and regenerates with probability 0.32 per step when below capacity; patch B yields 1.6 and regenerates with probability 0.55. Moving between patches costs 0.65. Each episode lasts 40 steps. These values are illustrative modeling assumptions, not measured biological facts.

MDP element	Definition
State s	(current patch, food in A, food in B): 2 × 4 × 4 = 32 states
Action a	0 = stay; 1 = switch to the other patch
Reward r	Harvest value at chosen patch (if food exists) − 0.65 if switching
Transition	Harvest 1 available food unit; each below-capacity patch may regenerate 1 unit stochastically
Objective	Maximize expected discounted cumulative reward over 40 steps

RL algorithm : Tabular Q-learning with learning rate $\alpha = 0.12$, discount $\gamma = 0.95$, and ε-greedy exploration decreasing from 1.0 to 0.05.

Update: $$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max Q(s',a') - Q(s,a)]$$

with zero future value at termination.

Baseline policies : Random: select stay/switch with equal probability. Heuristic: switch only when current patch is empty and the other patch contains food; otherwise stay. These policies provide simple reference points.

Hypothesis : Q-learning should exceed random selection after training, though it may not exceed the hand-designed heuristic.

IMPLEMENTATION AND REPRODUCICILITY

Implementation language : Python 3.

Dependencies: NumPy and Matplotlib. The full executable source is provided separately as ***foraging_qlearning.py***. No external dataset, internet access, or GPU is required.

Training consists of 2,500 episodes per independent random seed across 20 seeds (50,000 training episodes total). The agent uses a 32 × 2 Q-table, epsilon-greedy action selection, and online Q-learning updates. The epsilon schedule declines linearly for the first 1,800 episodes, then remains at 0.05.

Each final greedy policy is evaluated for 200 episodes per seed. All three methods are evaluated using the same environment seed for each training run. Because their actions differ, trajectories need not remain identical. Summary scores are means of 20 seed-level evaluation averages; reported SD is the variation across those 20 averages.

MEASURED RESULTS

Policy	Mean reward / episode	SD across seeds
Random	44.82	0.40
Heuristic	58.78	0.41
Q-learning	52.81	3.52

Q-learning exceeds the random baseline by **17.8%** in mean reward. It remains **5.96** reward units below the heuristic. Thus the experiment supports learning relative to random behavior but does not establish superiority over a simple hand-designed rule.

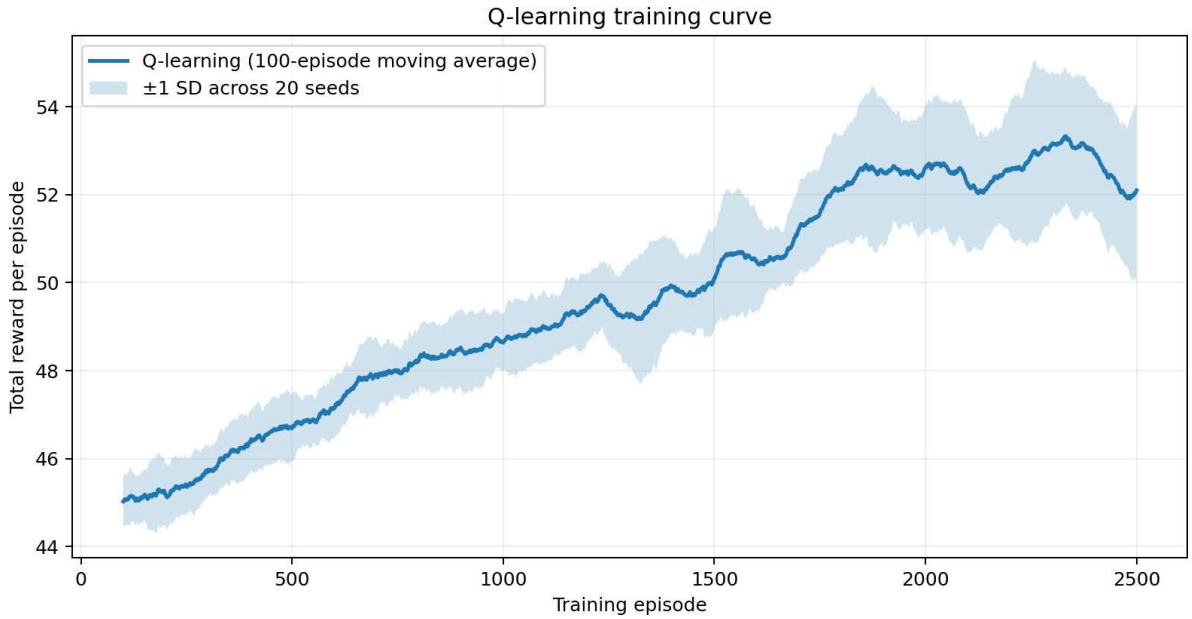


Figure 1. Training rewards smoothed with a 100-episode moving average; shaded band indicates ±1 SD across independent runs.

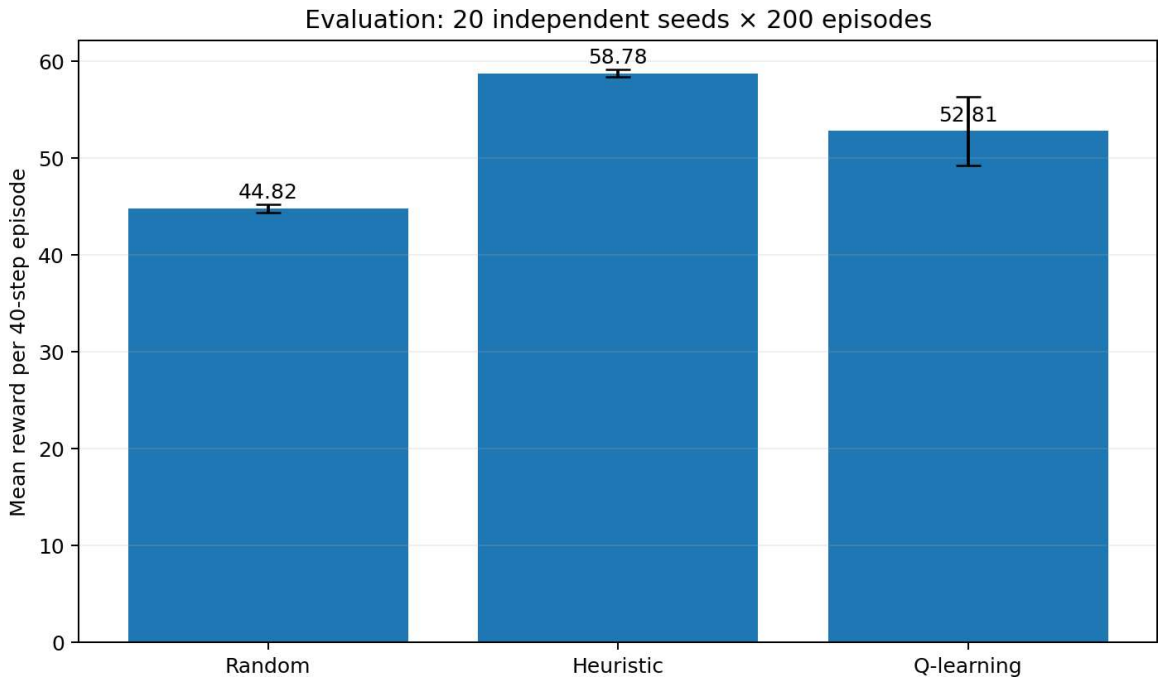


Figure 2. Evaluation reward for each policy; error bars indicate SD across 20 seed-level averages.

DISCUSSION AND LIMITATIONS

The agent discovers a useful policy without being given the heuristic rule. Nevertheless, Q-learning underperforms the heuristic in this particular small MDP. Possible causes include finite training, exploration during training, stochastic regeneration, and a strong domain-specific heuristic. Hyperparameter tuning or more training might change the outcome but has not been tested here.

This is a simplified simulation rather than a biological validation. Only two patches, discrete food quantities, fixed rewards, independent regeneration probabilities, and fully observed state are modeled. Real animals face uncertain perception, variable energy requirements, predators, and changing habitats. No conclusions about actual animal cognition can be drawn from this toy model.

ETHICS AND RESPONSIBLE INTERPRETATION

The experiment uses no animals or personal data and presents no direct animal-welfare risk. However, simulated decisions should not be misrepresented as real animal behavior. All model assumptions, random seeds, and limitations are documented to support reproducibility.

CONCLUSION

A tabular Q-learning agent was successfully trained to make sequential foraging decisions in a stochastic two-patch environment. It achieved higher average reward than random selection, but a simple heuristic was more effective. This demonstrates both the usefulness and the limitations of reinforcement learning in a compact simulated biological decision problem.